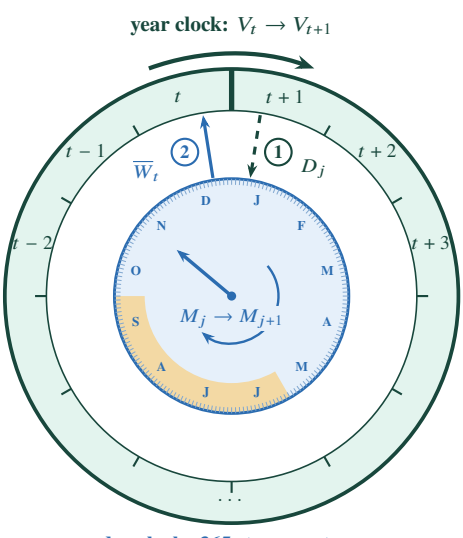


Year clock — cover, 1 step/yr

$$V_{t+1} = V_t + rV_t\left(1 - \frac{V_t}{K}\right)\overline{W}_t - dV_t - kE_t\left(1 - \eta\frac{V_t}{K}\right)V_t$$

V_t fixes demand D_j (1)

(2) $\overline{W}_t = \sum T_a / \sum D$

Day clock — water, 365 steps/yr

$$M_{j+1} = M_j + \alpha_P P_j + \eta_w I_j - T_{a,j} - Q_j$$

$$T_{a,j} = \min\left(M_j,\ K_s D_j\right), \quad D_j \propto f(V_t)\,\mathrm{ET}_{0,j}$$